



U-IMS

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University Issue Management System

United international university | DBMS project proposal

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check out my github repo <https://github.com/MahinMahbub>

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EXECUTIVE SUMMARY

Universities contain thousands of students who regularly face problems related to infrastructure, education, administration, transportation, sanitation, information technology, security, and other campus services. Examples include damaged classroom equipment, broken lights or fans, poor internet connectivity, water supply problems, problems with laboratory facilities, academic scheduling issues, transportation and other institutional concerns.

In many cases, students report these problems through informal channels such as directly contacting attendants, department offices, social media groups, or verbal complaints. Such methods can make it difficult to properly record, verify, prioritize, track, and resolve issues. Students may also have no convenient way to determine whether their complaint has actually reached the responsible authority.

U-ims is proposed as a centralized web-based University Issue Reporting and Resolution Management System. The system will allow students to submit university-related problems through a web interface, provide descriptions and supporting evidence such as photographs, and track the progress of their submitted issues.

Unlike a simple complaint form, the proposed system will introduce a **verification and publication process**. When a student submits an issue, an authorized university authority will review the submission and its evidence. The authority can verify, reject, categorize, prioritize, and assign the issue to the appropriate department or responsible person.

Verified issues can then appear on a public issue dashboard, allowing students to see which university problems have already been recognized and what their current status is.

The system will therefore create a structured communication channel between students and university authorities while maintaining a centralized database of university issues.

Recent studies have identified manual complaint handling as a source of delays, poor documentation, weak feedback mechanisms, and

reduced transparency. Web-based systems have been proposed to provide centralized complaint submission, status tracking, role-based access, and administrative reporting.

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PROBLEM STATEMENT

Students often encounter university problems but do not have a centralized and transparent platform through which they can report and monitor if those problems are being solved.

The existing informal approach creates several problems:

2.1 Lack of Centralized Reporting

Students may report different problems to different departments or individuals. As a result, there is no single database containing all reported university issues.

2.2 Problems May Be Forgotten or Lost

Verbal complaints or messages through informal communication channels may not be properly recorded. Consequently, some issues may remain unresolved.

2.3 Lack of Evidence

It can be difficult for authorities to verify whether a reported problem actually exists when the complaint contains only a text description.

2.4 Lack of Transparency

Students may submit a complaint without knowing:

- Whether the authority received it
- Whether it was verified
- Who is responsible for it
- Whether work has started
- Whether the problem has been resolved

2.5 Difficulty in Prioritizing Problems

Not all problems have the same level of importance. For example, a broken classroom fan and a dangerous electrical problem should not necessarily receive the same priority.

2.6 Difficulty Identifying Repeated Problems

If multiple students report the same issue independently, authorities may have difficulty identifying that these complaints refer to the same underlying problem.

2.7 Poor Historical Records

Without a centralized database, university authorities cannot easily analyze previous problems, resolution times, frequently affected locations, or departments with recurring issues.

Proposed Solution

U-ims will provide a centralized database-driven platform where:

Student → Reports Issue → Authority Reviews → Evidence Verified
→ Issue Published → Department Assigned → Issue Resolved →
Student Notified

This creates a structured lifecycle for university issue management.

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Benchmark Analysis / Literature Review

Several existing studies and systems demonstrate the usefulness of digital grievance and complaint-management platforms.

3.1 Web-Based Complaint Management Systems

A 2025 literature review of web-based complaint management systems identified common problems with manual and informal grievance processes, including delays, lack of transparency, and difficulty tracking outcomes. The reviewed system included complaint submission, evidence uploads, status tracking, role-based access, notifications, administrative dashboards, and analytics.

These features provide a useful benchmark for U-ims. However, U-ims will focus specifically on **university infrastructure, academic, and institutional issues**, rather than functioning as a generic complaint system.

3.2 No Student Complaint and Response Management

No study proposed a web-based Student Complaint and Response Management System using **HTML, CSS, JavaScript, PHP, and MySQL**, which is very similar to the technology stack proposed for this project.

U-ims extends this concept by introducing a stronger **evidence-based verification workflow** and a public dashboard of verified university issues.

3.3 Offline Student Complaint Systems

Research on offline student complaint systems has found that digital platforms can make complaints easier to coordinate, monitor, track, and resolve while helping institutions identify recurring problem areas.

3.4 Existing Institutional Grievance Systems

Existing institutional grievance platforms demonstrate the usefulness of an online workflow that manages a grievance from registration through resolution. For example, the Ucam's grievance system describes no online process designed to automate grievance management from complaint registration to closure.

3.5 Research Gap / Proposed Improvement

Although existing complaint-management systems provide online submission and tracking, U-ims will focus more specifically on **campus infrastructure and university-wide issue visibility**.

The proposed system will emphasize:

- Evidence-based issue reporting
- Authority verification
- Issue categorization
- Location-based reporting
- Duplicate/related issue identification
- Public display of verified problems
- Priority levels
- Department assignment
- Resolution progress
- Historical issue records
- Statistical reports

Therefore, the project combines the concepts of a **student grievance system, issue tracker, and university infrastructure monitoring platform**.

04 FEATURE LIST

4.1 Student Registration and Login

Students will create accounts using information such as:

- Student ID
- Name
- Email
- Department
- Batch
- Password

Students will then log in to access their dashboard.

Passwords will be securely hashed before being stored in the database.

4.2 Student Dashboard

After logging in, students will have access to a personal dashboard showing:

- Total issues submitted
- Pending issues
- Issues under review
- Verified issues
- Resolved issues
- Rejected issues

Students can also access their complete issue history.

4.3 Issue Submission

Students can submit a new university issue using a structured form.

The form can contain:

- Issue title
- Detailed description

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- Issue category
 - Location
 - Date/time observed
 - Priority suggestion
 - Supporting photograph/document
 - Optional additional information

Example:

Title: Broken Air Conditioner in Room 305

Category: Infrastructure

Location: Academic Building – Room 305

Description: The air conditioner has not been working for approximately one week.

4.4 Issue Categories

Issues will be divided into predefined categories.

For example:

Infrastructure

- Broken fan
- Damaged furniture
- Electrical problems
- Water leakage
- Broken classroom equipment
- Damaged roads/walkways

Education / Academic

- Class scheduling problems
- Missing laboratory equipment
- Course-related problems
- Examination-related issues
- Problems with academic facilities

IT / Technology

- Wi-Fi problems
- Computer laboratory problems

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- Projector problems
 - University portal problems

Sanitation

- Garbage
- Unclean washrooms
- Drainage problems
- Water supply problems

Security

- Broken gates
- Poor lighting
- Security concerns

Other

For problems that do not fit the predefined categories.

The categories can be stored in a separate SQL table rather than being hard-coded.

4.5 Evidence Upload

Students can attach evidence to support their issue.

Examples include:

- Photographs
- Screenshots

For example, a student reporting a broken classroom fan can upload a photograph of the damaged fan.

4.6 Authority Verification

This is one of the most important features of the project.

Submitted issues will initially have a status such as:

Pending Verification

An authorized authority can inspect:

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- Student's description
 - Location
 - Evidence
 - Category
 - Date submitted

The authority can then:

Verify

The issue is genuine and should be published.

Reject

The issue is invalid, inappropriate, duplicate, or unsupported.

Request More Information

The authority can ask the student to provide additional information.

This prevents false or misleading reports from immediately appearing on the public issue list.

4.7 Verified Issue Dashboard

After an authority verifies an issue, it becomes visible on the **Verified Issues** section.

Students can browse verified issues and see:

- Issue title
- Category
- Location
- Description
- Date reported
- Current status
- Priority
- Assigned department
- Resolution updates

This makes the system more transparent.

4.8 Issue Status Tracking

Every issue will have a lifecycle.

For example:

Pending → Under Review → Verified → Resolved → Closed

An issue may also be:

Rejected

Students can view the current status of their submitted issues.

4.9 Priority Management

Authorities can assign priority levels:

- Low
- Medium
- High
- Critical

For example:

Critical: Exposed electrical wiring

High: Major water leakage

Medium: Broken classroom projector

Low: Minor furniture damage

This allows authorities to focus on more urgent problems.

4.10 Issue Search and Filtering

Users can search and filter issues based on:

- Issue title
- Category
- Location
- Status
- Priority

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- Department
 - Date

For example:

Show all **Infrastructure** issues that are **Unresolved** in the **Main Academic Building**.

4.11 Student Notifications

Students can receive notifications when their issue:

- Is submitted
- Is verified
- Is rejected
- Is resolved

For a simple project, these notifications can be implemented as database records displayed inside the website.

4.12 Authority Dashboard

Authorities will have a separate dashboard containing:

- New issues awaiting verification
- Verified issues
- Issues in progress
- Critical issues
- Resolved issues
- Recently submitted issues
- Issues by category
- Issues by department

This gives university staff a centralized management interface.

4.13 Admin Management

An administrator can manage:

- Student accounts

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- Authority accounts
 - Departments
 - Issue categories
 - Locations
 - User roles
 - Issue records

The administrator will have higher privileges than normal authority users.

4.14 Reports and Statistics

The system can generate useful statistics from the database.

Examples:

- Total issues
- Resolved issues
- Unresolved issues
- Issues by category
- Issues by department
- Issues by location
- Monthly issue count
- Average resolution time
- Number of critical issues

For example:

Infrastructure: 42 issues
Academic: 18 issues
IT: 27 issues
Sanitation: 13 issues

This demonstrates the practical analytical use of a relational database.

4.15 Public Issue Board

A public-facing page can show verified issues without exposing private student information.

Example:

Verified University Issues

Issue	Location	Category	Priority	Status
Broken Classroom Fan	432	Infrastructure	Medium	In Progress
Library Wi-Fi Problem	Library	IT	High	Verified
Water Leakage	302B	Infrastructure	Critical	Resolved

The identity of the student who submitted the issue does not need to be displayed publicly.

4.16 Role-Based Access Control

The system will contain different types of users.

Student

Can:

- Submit issues
- Upload evidence
- Track own issues
- View verified issues
- Receive updates

Authority

Can:

- Review issues
- Verify/reject issues
- Assign departments
- Change priority
- Update status

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- Post progress updates

Administrator

Can:

- Manage users
- Manage departments
- Manage categories
- Manage system settings
- View all reports

This will demonstrate an important real-world database concept: **role-based access control**.

Suggested Database Structure

Ten-table relational schema



This database design provides multiple relationships between entities and gives the project substantial scope for demonstrating DBMS concepts such as **primary keys, foreign keys, normalization, joins, CRUD operations, aggregation, filtering, and relational integrity**.

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TECHNOLOGY USED

FRONTEND

- HTML5
- CSS3
- JavaScript

For user interface.

BACKEND

- PHP

Handles **server-side processing**, such as user authentication, form submissions, business logic, and communication with the database.

DATABASE

- MySQL

Stores and manages the application's **data**, such as user accounts, submitted issues, comments, and issue status.

DEVELOPMENT ENVIRONMENT

- XAMPP

Provides a **local server environment** for development. It includes Apache for running PHP and MySQL for managing the database.

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LIMITATIONS

Although U-ims will provide a functional issue-management platform, the proposed academic version will have several limitations.

6.1 Manual Verification

Issues will be verified manually by authorized personnel. The system will not automatically determine whether a submitted complaint is genuine.

6.2 No Mobile Application

The project will initially be a web application rather than a dedicated Android or iOS application.

6.3 File Storage Limitations

Uploaded photographs and documents will be stored on the local server during development. A production system would require more robust cloud or dedicated storage.

6.4 Security Limitations

The project will implement basic web-security practices such as password hashing, input validation, prepared SQL statements, session management, and access control. However, a real university deployment would require professional security auditing and additional protections.

6.5 Authority Response Depends on Humans

The system can track and notify authorities about problems, but it cannot physically repair infrastructure or force a department to resolve an issue.

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CONCLUSION

U-IMS: University Issue Management System is a database-driven web application designed to create a structured communication channel between university students and authorities.

The system will allow students to report infrastructure, academic, IT, sanitation, security, and other university-related problems with supporting evidence. Authorized university personnel will be able to review and verify these submissions before publishing them on the verified issue dashboard.